

Pedro Daniel Rocha

PhD Student – Electrical Engineering and Intelligent Systems

Pedro Daniel Rocha received the B.Sc. and M.Sc. degrees in Electrical Engineering from the Coimbra Institute of Engineering, Polytechnic University of Coimbra, respectively in 2021 and 2024. He is currently pursuing the Ph.D. degree in Electrical Engineering and Intelligent Systems at the University of Coimbra, Portugal. From 2012 to 2024, he worked across various manufacturing industries, holding roles ranging from technician to supervisor and development engineer. During this period, he was involved in projects related to computer vision technologies, automated systems, and IoT solutions. Since 2024, he has been a Researcher with the Instituto de Telecomunicações, Portugal. His research interests include computer vision applications, deep learning innovations, effective systems engineering, and data science and analytics.

EDUCATION

PhD in Electrical Engineering and Intelligent Systems ————— 2024 – Present

- Faculty of Sciences and Technology of the University of Coimbra
Specialization in Computer Vision and Machine Learning
Thesis Project : Automated Visual Inspection of Electrical Grid Assets using Active Learning and Embedded Vision Systems
Current Average Classification: **19/20**



Master's Degree in Electrical Engineering ————— 2021 – 2024

- Coimbra Institute of Engineering – Polytechnic University of Coimbra
Specialization in Automation and Communications in Industrial Systems
Thesis: Automated Visual Inspection of Electrical Grid Assets using DL
Final Average Classification: **18/20**



Bachelor's Degree in Electrical Engineering ————— 2018 – 2021

- Coimbra Institute of Engineering – Polytechnic University of Coimbra
Final Project: Micro-Mouse Robot Development
Final Average Classification: **16/20**



Technical Course in Electrotechnical and Electronics ————— 2007 – 2010

- E.S. Dr. Bernardino Machado – Figueira da Foz
Final Project : Line Follower Robot Development
Final Average Classification: **15/20**

PUBLICATIONS

ACCEPTED

P. D. Rocha, L. Cruz, I. Vilela, A. Coelho and F. Lopes, "Leveraging Error-Tolerance Asymmetry in Electrical Grid Automated Visual Inspection with a Semi-Supervised Annotation Pipeline", 2026 IEEE International Conference on Image Processing (ICIP), Tampere, Finland, 2026.

P. D. Rocha, A. Coelho, R. Santos, R. Martins, L. Cruz and F. Lopes, "Automating Electrical Grid Asset Inspections: Improving Insulators Health Assessment with Region-Level Annotations," in CIGRE Paris Session 2026.

PUBLISHED

P. D. Rocha, L. A. da Silva Cruz, A. Coelho and F. Lopes, "Automating Electrical Grid Asset Inspections Using Deep Vision: Evaluating the Impact of Annotation Detail Level", 2025 14th International Symposium on Image and Signal Processing and Analysis (ISPA), Coimbra, Portugal, 2025, pp. 199-203, doi: 10.1109/ISPA66905.2025.11259463.

P. D. Rocha, F. J. P. Lopes and L. A. Da Silva Cruz, "Automating Electrical Grid Asset Inspection: From Current Challenges to Future Directions", in IEEE Access, vol. 13, pp. 201392-201438, 2025, doi: 10.1109/ACCESS.2025.3636718.

P. D. Rocha, F. Lopes and L. A. Da Silva Cruz, "Advances in Electrical Grid Assets Inspection: Exploring Multimodal Large Language Models", 2025 25th International Conference on Digital Signal Processing (DSP), Pylos (Messinia, Southwest Peloponnese), Greece, 2025, pp. 1-5, doi: 10.1109/DSP65409.2025.11075122.

F. Lopes, **P. Rocha** and A. Coelho, "Towards Automated Visual Inspection of Electrical Grid Assets for the Smart Grid - An Application to HV Insulators", 2024 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), Oslo, Norway, 2024, pp. 347-352, doi: 10.1109/SmartGridComm60555.2024.10738034.



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MEMBERSHIPS

IEEE – Graduate Student Member

Nº 100595034



CIGRE – Student Member

Nº 920251250



RESEARCH EXPERIENCE



Instituto de Telecomunicações – Coimbra, Portugal

Doctoral Researcher (PhD Scholarship) ————— 2025 – Present

- Collaborate with Industrial Partners for Understanding Operational Requirements
- Explore and Develop Advanced Deep Learning Architectures and Algorithms
- Document and Publish Research Methodologies and Performance Results

Graduate Researcher (MSc Research Grant) ————— 2024 – 2025

- Literature Review and Identification of Current Challenges in the Field
- Analyze and Curate Annotated Datasets to Train and Test CV Models

TEACHING EXPERIENCE



University of Coimbra – Coimbra, Portugal

Teaching Assistant ————— 2025 – Present

- Digital Signal Processing
- Analogic Electronics II

INDUSTRIAL EXPERIENCE



EDP Labelec – Lisbon, Portugal

Data Scientist (PhD Internship) ————— 2025 – Present

- Collaborate in the development and integration of data annotation tools.
- Develop pipelines for dataset collection and preprocessing to support ML workflows.
- Contribute to data acquisition optimization strategies for real-world inspection applications.



CWJ Power Electronics, SA – Figueira da Foz, Portugal

Embedded Systems Engineer ————— 2023 – 2024

- Analyze System Concept and Document System Requirements
- Propose Architectural and System Design Definitions
- Promote System Integration and Verification



Verallia Portugal, SA, SA – Figueira da Foz, Portugal

Electro-Mechanical Maintenance Supervisor ————— 2022 – 2023

- Lead Maintenance Technician Crew
- Develop and Implement Maintenance Plans
- Organize and Manage an Equipment Park

Instrumentation and Automation Technician ————— 2019 – 2022

- Diagnose and Solve Breakdowns in Automatic Industrial Machines
- Debug PLC Programs in Automatic Machines and Production Lines
- Parameterize and Fine Tune Visual Inspection Machines

Production Operator ————— 2018 – 2019

- Operate Automatic Industrial Machines
- Control Finished Product Quality

FozMel – Montagens Eléctricas, Lda– Figueira da Foz, Portugal

Industrial Electrician ————— 2011 – 2018

- Build Electrical Switch Boards according to the Project's Electrical Plan
- Identify and Locate Faults in Power and Command Circuits

ACADEMIC SERVICE

Peer Reviewer ————— 2025 – Present

- IEEE International Conference on Image Processing (ICIP 2026)
- Open Journal of Signal Processing (OJSP)
- Signal Processing: Image Communication

Local Organizing Committee Member ————— 2025

- International Symposium on Image and Signal Processing and Analysis
- 29-31 October 2025, Coimbra, Portugal
- <https://www.isispa.org/>

HONORS AND AWARDS

Runner-Up, Best Student Paper Award

- International Symposium on Image and Signal Processing and Analysis (ISPA 2025)
- 29-31 October 2025, Coimbra, Portugal
- <https://www.isispa.org/>